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Previous Year Paper
Mechanical Engineer
04 Nov 2022 Shift 2



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Participant ID	
Participant Name	
Test Center Name	
Test Date	04/11/2022
Test Time	2:00 PM - 4:30 PM
Subject	MECHANICAL ENGINEER

Section : English Language

Q.1 Select the most appropriate antonym of the given word.

IRATE

- Ans
- ☒ 1. furious
 - ☒ 2. annoyed
 - ☒ 3. cheerful
 - ☒ 4. indignant

Question ID : 8401605732
Status : Answered
Chosen Option : 3

Q.2 Select the most appropriate option to fill in the blank.

In 1025 AD, Mahmud Gazni _____ an attack on the Somnath temple in order _____ the wealth that had gathered within the temple.

- Ans
- ☒ 1. to launch ; stealing
 - ☒ 2. launch ; stole
 - ☒ 3. launching ; steal
 - ☒ 4. launched ; to steal

Question ID : 8401605742
Status : Answered
Chosen Option : 4

Q.3 Select the most appropriate option to fill in the blank.

Sanjana is _____ different from her mother.

- Ans
- ☒ 1. a very
 - ☒ 2. very
 - ☒ 3. the very
 - ☒ 4. much very

Question ID : 8401605747
Status : Answered
Chosen Option : 2



Q.4 Select the most appropriate meaning of the given idiom.

put your foot down

- Ans ☒ 1. to be firm about something
- ☐ 2. to squash an insect
- ☐ 3. to make great effort
- ☐ 4. to get into trouble

Question ID : 8401605737
Status : Answered
Chosen Option : 3

Q.5 Select the most appropriate option to fill in the blank.

One pirate ship was _____ by the king' s soldiers but the other one _____.

- Ans ☐ 1. kidnapped ; deserted
- ☐ 2. abducted ; arived
- ☒ 3. captured ; escaped
- ☐ 4. collected ; broke

Question ID : 8401605741
Status : Answered
Chosen Option : 3

Q.6 Given below are four jumbled sentences. Select the option that gives their correct order forming a meaningful and coherent paragraph.

- A. If the Beatrice is ever found, it may be possible to retrieve the ancient sarcophagus.
- B. In the 1830s, English military officer Howard Vyse explored the Giza pyramids.
- C. Vyse tried to ship the sarcophagus to England in 1838 aboard the merchant ship Beatrice, but the ship sank during its journey and took the ornate sarcophagus down with it.
- D. Vyse made a number of discoveries at Giza, including an ornate sarcophagus found in Menkaure's pyramid.

- Ans ☐ 1. ADBC
- ☐ 2. BACD
- ☒ 3. BDCA
- ☐ 4. CDAB

Question ID : 8401605740
Status : Answered
Chosen Option : 3

Q.7 Select the most appropriate option to fill in the blank.

This time next week, they _____ in Canada.

- Ans ☐ 1. have been
- ☐ 2. be
- ☒ 3. will be
- ☐ 4. had been

Question ID : 8401605743
Status : Answered
Chosen Option : 3

Q.8 Select the most appropriate option to fill in the blank.

People were quite prepared that day _____ the demolition of the Supertech Twin Towers in NOIDA.

- Ans
- ☒ 1. at
 - ☒ 2. of
 - ☒ 3. for
 - ☒ 4. to

Question ID : 8401605745
Status : Answered
Chosen Option : 3

Q.9 Select the most appropriate synonym of the given word.

PENURIOUS

- Ans
- ☒ 1. economical
 - ☒ 2. benevolent
 - ☒ 3. extravagant
 - ☒ 4. prodigal

Question ID : 8401605735
Status : Answered
Chosen Option : 3

Q.10 Select the most appropriate option to fill in the blank.

My brother Anurag is one year _____ than your brother Rohit.

- Ans
- ☒ 1. oldest
 - ☒ 2. older
 - ☒ 3. elder
 - ☒ 4. eldest

Question ID : 8401605746
Status : Answered
Chosen Option : 3

Q.11 Select the most appropriate meaning of the given idiom.

rock the boat

- Ans
- ☒ 1. escape from problems
 - ☒ 2. act without thinking
 - ☒ 3. disturb an existing situation
 - ☒ 4. enjoy a boat ride

Question ID : 8401605736
Status : Answered
Chosen Option : 1



Q.12 Select the option which is NOT an antonym of another word by way of adding the prefix 'in- '.

- Ans ☒ 1. innumerable
☒ 2. inoffensive
☒ 3. insidious
☒ 4. inscrutable

Question ID : 8401605733
 Status : Answered
 Chosen Option : 3

Q.13 Select the most appropriate option to fill in the blank.

She _____ her homework yet; she cannot go out to play.

- Ans ☒ 1. does not finish
☒ 2. did not finish
☒ 3. has not finished
☒ 4. is not finishing

Question ID : 8401605744
 Status : Answered
 Chosen Option : 3

Q.14 Select the correct option to fill in the blanks.

Throughout history, fantastic treasures from various cultures have been stolen or _____ gone missing.

- Ans ☒ 1. misteriously
☒ 2. mysiterously
☒ 3. mystiriously
☒ 4. mysteriously

Question ID : 8401605738
 Status : Answered
 Chosen Option : 4

Q.15 Select the most appropriate synonym of the given word.

TENDER

- Ans ☒ 1. sturdy
☒ 2. robust
☒ 3. durable
☒ 4. delicate

Question ID : 8401605734
 Status : Answered
 Chosen Option : 4

Q.16 Select the segment which has a spelling error in the given sentence. If there is no error, select 'No error'.

The National Education Policy, 2020, is based on the foundational principals of access, equity, quality, affordability and accountability..

- Ans
- ☒ 1. of access, equity, quality, affordability and accountability..
 - ☒ 2. The National Education Policy, 2020
 - ☒ 3. No error
 - ☒ 4. is based on the foundational principals

Question ID : 8401605739
Status : Answered
Chosen Option : 3

Q.17 Select the option that completes the given proverb correctly.

_____ of all trades is master of none.

- Ans
- ☒ 1. A speaker
 - ☒ 2. A jack
 - ☒ 3. A king
 - ☒ 4. A workman

Question ID : 8401605748
Status : Answered
Chosen Option : 2

Section : Quantitative Aptitude

Q.1 The vulgar fraction of $0.\overline{32} + 0.2\overline{6} - 0.\overline{53}$ is:

- Ans
- ☒ 1. $\frac{27}{495}$
 - ☒ 2. $\frac{27}{990}$
 - ☒ 3. $\frac{54}{495}$
 - ☒ 4. $\frac{53}{495}$



Question ID : 8401605752
Status : Not Answered
Chosen Option : --



Q.2 Two marbles are drawn in succession from a box containing 10 red, 30 white, 20 blue and 15 orange marbles. Find the probability that the first drawn marble is red and the second drawn marble is white.

- Ans
- ☒ 1. $\frac{8}{75}$
 - ☒ 2. $\frac{9}{75}$
 - ☒ 3. $\frac{4}{75}$
 - ☒ 4. $\frac{6}{75}$

Question ID : 8401605777
Status : Answered
Chosen Option : 2

Q.3 The largest length of tape to be used to measure a room' s sides, whose distances are 6 m 48 cm, 11m 52 cm and 16m 20 cm is:

- Ans
- ☒ 1. 36
 - ☒ 2. 72
 - ☒ 3. 45
 - ☒ 4. 32

Question ID : 8401605751
Status : Answered
Chosen Option : 2

Q.4 A vendor earns 25% profit on selling his goods at Rs.2,355. The cost price of the goods is:

- Ans
- ☒ 1. Rs.1,488
 - ☒ 2. Rs.1,848
 - ☒ 3. Rs.1,884
 - ☒ 4. Rs.1,484

Question ID : 8401605760
Status : Answered
Chosen Option : 3

Q.5 The diagonal of a cube is $6\sqrt{3}$ ft. If 1 kg paint covers 12 sq ft, how much will it cost to paint the exterior of the cube if the cost of paint per kg is Rs.320?

- Ans
- ☒ 1. Rs.5,760
 - ☒ 2. Rs.5,670
 - ☒ 3. Rs.5,706
 - ☒ 4. Rs.5,607

Question ID : 8401605780
Status : Answered
Chosen Option : 1



Q.6 If two numbers are in the ratio 3 : 5 and their LCM is 45, then the sum of those numbers is:

- Ans
- ☐ 1. 22
 - ☐ 2. 9
 - ☐ 3. 15
 - ☒ 4. 24

Question ID : 8401605750
Status : Answered
Chosen Option : 4

Q.7 The monthly salary of teaching and non-teaching staff at a college is in the ratio of 5 : 3. They wish to contribute 3% and 2% of their salaries to a welfare society. If each non-teaching staff member contributes Rs.390, then the teaching staff salary is:

- Ans
- ☒ 1. Rs.32,500
 - ☐ 2. Rs.32,050
 - ☐ 3. Rs.35,200
 - ☐ 4. Rs.35,020

Question ID : 8401605765
Status : Answered
Chosen Option : 2

Q.8 The length of train A is 100 m more than the length of a platform. What is the time taken by train A to cross train B travelling in the opposite direction if the speed of train B is 70 km/h and the speed of train A is 90 km/h and the time taken by train A and B to cross the same platform is 24 seconds and 36 seconds, respectively?

- Ans
- ☐ 1. 16 seconds
 - ☒ 2. 18 seconds
 - ☐ 3. 17 seconds
 - ☐ 4. 15 seconds

Question ID : 8401605770
Status : Answered
Chosen Option : 2

Q.9 The distance between two stations, Mathura and Haridwar, is 356 km. Train ABC Express, which is running at a speed of 72 km/h, leaves Mathura station at 10:50 p.m. Train XYZ Express, which is running at the speed of 82 km/h, leaves Haridwar station at 11:30 p.m. At what time will ABC Express and XYZ Express meet each other?

- Ans
- ☐ 1. 1:50 a.m.
 - ☐ 2. 1:00 a.m.
 - ☐ 3. 12:50 a.m.
 - ☒ 4. 1:30 a.m.

Question ID : 8401605771
Status : Answered
Chosen Option : 4

Q.10 A dealer used a faulty weighing machine and measured 950 g instead of 1 kg and claims he sell his goods at cost price. His profit percentage is:

- Ans
- ☒ 1. 4.26%
 - ☒ 2. 4.56%
 - ☒ 3. 5.26%
 - ☒ 4. 5.56%

Question ID : 8401605761
Status : Answered
Chosen Option : 2

Q.11 What annual installment will discharge a debt of Rs.1,431 due in three years at 6% simple interest?

- Ans
- ☒ 1. Rs. 445
 - ☒ 2. Rs. 350
 - ☒ 3. Rs. 550
 - ☒ 4. Rs. 450

Question ID : 8401605767
Status : Answered
Chosen Option : 4

Q.12 Ram and Syam are working in an IT company with salaries of Rs.23,500 and Rs.32,500, respectively. The chairman of the company wishes to give increments of 6% proportionate to their salaries. Their new salaries will be in the ratio of:

- Ans
- ☒ 1. $\frac{47}{65}$
 - ☒ 2. $\frac{45}{67}$
 - ☒ 3. $\frac{67}{45}$
 - ☒ 4. $\frac{65}{47}$

Question ID : 8401605766
Status : Answered
Chosen Option : 2

Q.13 A' s income is two-fifth of B' s income. The expenditure of A is 50% of B' s expenditure and the income, expenditures of B are Rs. 48250, 28% of the income of A, respectively. Then the savings of A is:

- Ans
- ☒ 1. Rs.16,598
 - ☒ 2. Rs.16,985
 - ☒ 3. Rs.16,589
 - ☒ 4. Rs.16,895

Question ID : 8401605756
Status : Answered
Chosen Option : 2



Q.14 The given chart shows the increase in weight of a boy over the given years. Study the given chart and answer the question that follows.



What is the percentage change in weight of the boy from 2018 to 2019?

- Ans
- ☒ 1. 80%
 - ☒ 2. 75%
 - ☒ 3. 25%
 - ☒ 4. 50%

Question ID : 8401605772
Status : Answered
Chosen Option : 2

Q.15 If the curved surface area of a right circular cylinder is 3696 cm^2 and the circumference of its base is 88 cm , then find its volume.

- Ans
- ☒ 1. 26256 cm^3
 - ☒ 2. 25872 cm^3
 - ☒ 3. 27318 cm^3
 - ☒ 4. 24524 cm^3

Question ID : 8401605781
Status : Answered
Chosen Option : 2

Q.16 In what ratio must water be mixed with milk to gain 20% by selling the mixture at cost price?

- Ans
- ☒ 1. 2 : 3
 - ☒ 2. 1 : 5
 - ☒ 3. 3 : 4
 - ☒ 4. 2 : 5

Question ID : 8401605755
Status : Answered
Chosen Option : 1

Q.17 Two bowlers A and B take average wickets of 28 and 34 in a series of 8 and 12 matches, respectively. If they took 5 wickets in the 9th match and 1 wicket in the 13th match, then the average number of wickets of both the bowlers is:

- Ans
- ☒ 1. 30
 - ☒ 2. 29
 - ☒ 3. 28
 - ☒ 4. 32

Question ID : 8401605754
Status : Answered
Chosen Option : 1

Q.18 Krishna's salary was increased by 8% in the first year and then increased by 9% in the next year. In the third year, he earned Rs.2,536 for working overtime. If his initial salary was Rs.32,500, his total salary drawn in the third year is:

- Ans
- ☒ 1. Rs.40,975
 - ☒ 2. Rs.40,795
 - ☒ 3. Rs.40,579
 - ☒ 4. Rs.40,979

Question ID : 8401605757
Status : Answered
Chosen Option : 2

Q.19 The factorisation of the equation $2x^2 + 7x + 6 = 0$ is:

- Ans
- ☒ 1. $(2x - 3)(x + 2) = 0$
 - ☒ 2. $(2x + 3)(x + 2) = 0$
 - ☒ 3. $(2x + 3)(x - 2) = 0$
 - ☒ 4. $(2x - 3)(x - 2) = 0$

Question ID : 8401605779
Status : Answered
Chosen Option : 2

Q.20 The ratio of the percentage $16\frac{2}{3}\%$ is:

- Ans
- ☒ 1. $\frac{1}{8}$
 - ☒ 2. $\frac{1}{5}$
 - ☒ 3. $\frac{1}{6}$
 - ☒ 4. $\frac{1}{7}$

Question ID : 8401605758
Status : Answered
Chosen Option : 3

Q.21 A, B and C are trains travelling from Delhi to Ahmedabad at speeds of 45 km/h, 55 km/h and 61 km/h in 3 hours, 2 hours and 1 hour, respectively. Their average speed is:

- Ans
- ☒ 1. 55 km/h
 - ☒ 2. 45 km/h
 - ☒ 3. 51 km/h
 - ☒ 4. 41 km/h

Question ID : 8401605769
Status : Answered
Chosen Option : 2

Q.22 A dealer has the following three schemes running for products in his store. Which of the following has maximum discount percentage?

- I. Two successive discounts of 15% and 20%
- II. Buy 5 get 3
- III. Buy 5 get 6

- Ans
- ☒ 1. Only I and III
 - ☒ 2. Only I
 - ☒ 3. Only III
 - ☒ 4. Only II

Question ID : 8401605764
Status : Answered
Chosen Option : 1

Q.23 The value of $\sqrt{14 + 2\sqrt{45}}$ is:

- Ans
- ☒ 1. $\sqrt{3} + \sqrt{5}$
 - ☒ 2. $3 - \sqrt{5}$
 - ☒ 3. $\sqrt{3} - \sqrt{5}$
 - ☒ 4. $3 + \sqrt{5}$

Question ID : 8401605749
Status : Answered
Chosen Option : 3

Q.24 If the area of a circular field is 3850 m^2 , then the cost of fencing it at the rate of Rs.52 per metre is:

- Ans
- ☒ 1. Rs.11,440
 - ☒ 2. Rs.11,230
 - ☒ 3. Rs.11,220
 - ☒ 4. Rs.11,560

Question ID : 8401605782
Status : Answered
Chosen Option : 2



Q.25 The number of 5-digit numbers that can be formed by the digits 2, 2, 3, 3 and 4 is:

- Ans
- ☒ 1. 35
 - ☒ 2. 25
 - ☒ 3. 20
 - ☒ 4. 30

Question ID : 8401605774
Status : Answered
Chosen Option : 2

Q.26 There are two number in the ration 5 : 3. If the difference between the two numbers is 22, find the larger number.

- Ans
- ☒ 1. 55
 - ☒ 2. 33
 - ☒ 3. 44
 - ☒ 4. 22

Question ID : 8401605776
Status : Answered
Chosen Option : 1

Q.27 A merchandiser sells bananas and apples at a gain of 20% on bananas and at a loss of 5% on apples. If the cost price of both fruits is Rs.5,000 and he earned 6% on the whole, then the cost price of bananas is:

- Ans
- ☒ 1. Rs.2,200
 - ☒ 2. Rs.2,800
 - ☒ 3. Rs.2,820
 - ☒ 4. Rs.2,500

Question ID : 8401605759
Status : Answered
Chosen Option : 1

Q.28 A family wishes to attend a party 150 km away from their home. They travel at an average speed of 50 km/h and return at a speed of 75 km/h. Their average speed for the whole journey is:

- Ans
- ☒ 1. 60 km/h
 - ☒ 2. 65 km/h
 - ☒ 3. 57 km/h
 - ☒ 4. 62 km/h

Question ID : 8401605753
Status : Answered
Chosen Option : 1

Q.29 What is the nature of the roots of $3x^2 - 6x + 5 = 0$?

- Ans
- ☒ 1. The roots are real and more than 2.
 - ☒ 2. The roots are real and distinct.
 - ☒ 3. The roots are real and equal.
 - ☒ 4. There are no real roots.

Question ID : 8401605778
Status : Answered
Chosen Option : 3

Q.30 What annual instalment will discharge a debt of Rs.16550 due in three years at 10% compound interest annually?

- Ans
- ☒ 1. Rs. 6,556
 - ☒ 2. Rs. 5,665
 - ☒ 3. Rs. 6,655
 - ☒ 4. Rs. 5,566

Question ID : 8401605768
Status : Answered
Chosen Option : 2

Q.31 An automobile accessory costs Rs.5,825. A customer requested a discount of 18%, in addition to the discount of 5% already offered by the merchant. The cost of the accessory after these discounts will be:

- Ans
- ☒ 1. Rs. 4,547.675
 - ☒ 2. Rs. 4,536.675
 - ☒ 3. Rs. 4,537.675
 - ☒ 4. Rs. 4,637.675

Question ID : 8401605763
Status : Answered
Chosen Option : 2

Q.32 A merchant marked his goods at 20% above the cost price and offered a discount of 5%. Seeing poor results, he again offered 8% discount and noticed a growth in his sales. His profit will be:

- Ans
- ☒ 1. 10.88%
 - ☒ 2. 4.88%
 - ☒ 3. 3.88%
 - ☒ 4. 5.88%

Question ID : 8401605762
Status : Answered
Chosen Option : 2

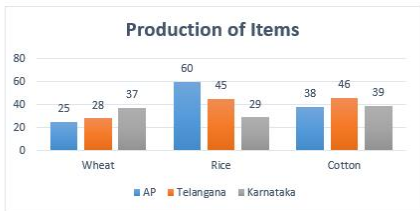


Q.33 Ten years ago, a man is six times as old as his daughter. Three years hence, thrice his age will be equal to five times of his daughter's age. The present age of his daughter is:

- Ans ☒ 1. 12 years
☒ 2. 14 years
☒ 3. 13 years
☒ 4. 15 years

Question ID : 8401605775
 Status : Answered
 Chosen Option : 1

Q.34 Study the given chart and answer the question that follows.



Find the ratio between the difference of rice in AP and Telangana to that of the difference of wheat in Karnataka and AP.

- Ans ☒ 1. 3 : 2
☒ 2. 5 : 4
☒ 3. 4 : 5
☒ 4. 2 : 3

Question ID : 8401605773
 Status : Answered
 Chosen Option : 2

Section : Intellectual Potential Test

Q.1 In the following series of alphabet identify the letter pattern and fill the blank in the series.

ZLM, YPQ, XST, WDE,

- Ans ☒ 1. UAB
☒ 2. VAB
☒ 3. UTP
☒ 4. VTP

Question ID : 8401605797
 Status : Answered
 Chosen Option : 2

Q.2 Amit is standing by facing East and took a turn of 90 degree in the anti-clockwise direction and then he turns another 180 degrees in the same direction and then 90 degree in the clockwise direction. Find which direction Amit is facing now ?

- Ans ☒ 1. East
☒ 2. West
☒ 3. North
☒ 4. South

Question ID : 8401605801
 Status : Answered
 Chosen Option : 1

Q.3 NATION:NOITAN::ACTION:?

- Ans ☒ 1. NOIACT
☒ 2. TCAION
☒ 3. NOTICE
☒ 4. NOITCA

Question ID : 8401605807
 Status : Answered
 Chosen Option : 4

Q.4 Read the following words and arrange their numbers by observing a pattern.

1. Baby 2. Adult 3. Child 4. Elder 5. Teenager

- Ans ☒ 1. 2,3,4,5,1
☒ 2. 1,3,4,5,2
☒ 3. 4,5,3,1,2
☒ 4. 1,3,5,2,4

Question ID : 8401605796
 Status : Answered
 Chosen Option : 4

Q.5 Statements: $B \leq C < A \geq D$, $E > F \geq D$

Conclusions: I. $E \geq A$ II. $F \leq C$ III. $A > B$ IV. $E \geq B$

Which one of the above conclusions is correct?

- Ans ☒ 1. Only III is true
☒ 2. Only III and IV are true
☒ 3. Only I and IV are true
☒ 4. All I, II, III and IV are true

Question ID : 8401605790
 Status : Answered
 Chosen Option : 2



Q.6 Uncertain number of people are sitting in a row by facing the North direction. R is sitting at one of the ends and there are two people between R and Q. Equal number of persons are sitting between U and R and U and S. Two persons are sitting between S and V. Equal number of persons are sitting to the right and left side of V. T is sitting exactly in the middle of P and S. U is third to the left of Q who is sitting fourth from one of the extreme ends of the row. P is sitting at one of the extreme ends of the row. U is not sitting on any of the extreme ends of the row. Number of people sitting between P and S are five.

Who is sitting between U and S?

- Ans
- ☒ 1. P
 - ☒ 2. Q
 - ☒ 3. R
 - ☒ 4. S

Question ID : 8401605788
Status : Answered
Chosen Option : 3

Q.7 Read the following numbers carefully and answer the questions as per direction given after the number series:

289 496 337 268 245

If all the digits are to be arranged in increasing order from left to right within the number, then what will be the difference between the highest number and the lowest number thus obtained?

- Ans
- ☒ 1. 422
 - ☒ 2. 442
 - ☒ 3. 224
 - ☒ 4. 242

Question ID : 8401605793
Status : Answered
Chosen Option : 2

Q.8 In the given question, pairs of words are given in which the words of any three pairs have a common relationship. Choose the pair of words which does not have that relationship.

- Ans
- ☒ 1. Boat: Water
 - ☒ 2. Car: Road
 - ☒ 3. Ship: Captain
 - ☒ 4. Train: Railway Track

Question ID : 8401605809
Status : Answered
Chosen Option : 3

Q.9 In the given question, set of pairs of alphabets are given out of which three pairs have a common relationship. Choose the pair of which does not have that same relationship.

- Ans ☒ 1. PUT: SXW
☒ 2. NOT: QRW
☒ 3. CAT: DBU
☒ 4. GOT: JRW

Question ID : 8401605810
 Status : Answered
 Chosen Option : 4

Q.10 Following table gives the details of sales of different items sold by three different stores run by three women. Please read the table carefully and answer the question given at the end of table.

Type / Sale in Thousand	Ambika Stores	Mahima Stores	Rohini Stores
Watches	312	234	453
Calculators	231	211	342
Mob. Phones	456	765	889
Coffee Mugs	2109	2306	3212

Find the difference of highest and lowest sale of coffee Mugs.

- Ans ☒ 1. 1303
☒ 2. 1003
☒ 3. 1203
☒ 4. 1103

Question ID : 8401605813
 Status : Answered
 Chosen Option : 4

Q.11 Pointing towards a woman, Sumit told to Garima that she is mother of only daughter of my son. What is relationship between the pointed woman and Sumit?

- Ans ☒ 1. Wife
☒ 2. Mother-in-law
☒ 3. Daughter-in-law
☒ 4. Mother

Question ID : 8401605805
 Status : Answered
 Chosen Option : 2



Q.12 Following table gives the details of sales of different items sold by three different stores run by three women. Please read the table carefully and answer the question given at the end of table.

Type / Sale in Thousand	Ambika Stores	Mahima Stores	Rohini Stores
Watches	312	234	453
Calculators	231	211	342
Mob. Phones	456	765	889
Coffee Mugs	2109	2306	3212

Find the total number of Mobile Phones sold by all the three Stores.

- Ans**
- ☒ 1. 2112
 - ☒ 2. 2113
 - ☒ 3. 2110
 - ☒ 4. 2111

Question ID : 8401605815
 Status : Answered
 Chosen Option : 3

Q.13 Following table gives the details of sales of different items sold by three different stores run by three women. Please read the table carefully and answer the question given at the end of table.

Type / Sale in Thousand	Ambika Stores	Mahima Stores	Rohini Stores
Watches	312	234	453
Calculators	231	211	342
Mob. Phones	456	765	889
Coffee Mugs	2109	2306	3212

Find the difference of total number of Calculators sold by all the Rohini and Ambika Stores.

- Ans**
- ☒ 1. 112
 - ☒ 2. 101
 - ☒ 3. 211
 - ☒ 4. 111

Question ID : 8401605816
 Status : Answered
 Chosen Option : 2

Q.14 A, B, C, D, E, F, G & H are eight friends sitting around a circular table by facing the centre. Three girls are sitting between A and D. A is sitting second to the left of B. H is not an immediate of B. F is sitting at the immediate left of C. D is not an immediate neighbor of either F or E.

Who is sitting at the immediate right of C?

- Ans**
- ☒ 1. C
 - ☒ 2. A
 - ☒ 3. D
 - ☒ 4. B

Question ID : 8401605784
 Status : Answered
 Chosen Option : 3

Q.15 Anuj told to Mohit that the boy Golu who is playing in the ground is the younger among the two brothers of the daughter of my father's wife. What is relationship between the boy playing in the ground with Anuj?

- Ans
- ☒ 1. Father
 - ☒ 2. Grandfather
 - ☒ 3. Brother
 - ☒ 4. Uncle

Question ID : 8401605804
Status : Answered
Chosen Option : 2

Q.16 Read the series: 8, 13, 11, 16, 14, 19, ... carefully and suggest the number that should come next?

- Ans
- ☒ 1. 21
 - ☒ 2. 24
 - ☒ 3. 22
 - ☒ 4. 17

Question ID : 8401605795
Status : Answered
Chosen Option : 4

Q.17 In the morning assembly of a school, in the single line of ninth class, Madan is standing at 28th place from both the ends of line. How many students are there in the line?

- Ans
- ☒ 1. 56
 - ☒ 2. 54
 - ☒ 3. 55
 - ☒ 4. 57

Question ID : 8401605785
Status : Answered
Chosen Option : 3

Q.18



In the above given Venn diagram rectangle represents teachers, triangle artists and circle sports persons. Give the number of the area represented by the all the three.

- Ans
- ☒ 1. 6
 - ☒ 2. 5
 - ☒ 3. 3
 - ☒ 4. 4

Question ID : 8401605789
Status : Answered
Chosen Option : 1

Q.19 Statement: $T = U \leq S < Q = P > R$

Conclusions: 1) $T > R$ 2) $P > T$ 3) $P < U$ 4) $R > S$

Which one of the above conclusions is correct?

- Ans
- ☒ 1. Only 1
 - ☒ 2. Only 4
 - ☒ 3. Only 2 & 4
 - ☒ 4. Only 2

Question ID : 8401605791
Status : Answered
Chosen Option : 2

Q.20 In a code sign MOHAN is coded as 13158114 and ROHIT is coded as 18158920 the what will be the code for ANUJ?

- Ans
- ☒ 1. 1118210
 - ☒ 2. 1182110
 - ☒ 3. 1042111
 - ☒ 4. 1142110

Question ID : 8401605799
Status : Answered
Chosen Option : 2

Q.21 If '+' means subtraction, '÷' means addition, '-' means multiplication and '×' means division, then which of the following equation is correct?

- Ans
- ☒ 1. $18 \div 128 - 4 \times 12 = 108$
 - ☒ 2. $112 + 24 \times 68 - 24 = 204$
 - ☒ 3. $224 \times 88 - 24 + 20 = 92$
 - ☒ 4. $16 \div 88 - 10 + 50 = 846$

Question ID : 8401605812
Status : Answered
Chosen Option : 2

Q.22 If B is sister of L and A is Mother of B and also L is father of K then what is relation of A with K?

- Ans
- ☒ 1. Mother
 - ☒ 2. Grandmother
 - ☒ 3. Daughter
 - ☒ 4. Sister

Question ID : 8401605806
Status : Answered
Chosen Option : 2

Q.23 If + stands for 'x', x stands for '+', - stands for '÷' and ÷ stands for '-', then
 $10 \times 4 + 8 - 4 + 12 - 6 \div 2 = ?$

- Ans
- ☒ 1. 42
 - ☒ 2. 26
 - ☒ 3. 24
 - ☒ 4. 32

Question ID : 8401605811
 Status : Answered
 Chosen Option : 2

Q.24 Two friends Ashish and Mohit start moving in the opposite direction on a main road and both are 1500 meters apart from each other. Ashish walks for 250 meters on the main road and takes a right turn and then walks for another 150 meters. Then he turns left and walks for another 250 meters and then turns in the direction to reach back to the main road. Meanwhile, Mohit could walk only 350 meters on the main road. What is the distance between both of them at this point?

- Ans
- ☒ 1. 650 M
 - ☒ 2. 450 M
 - ☒ 3. 550 M
 - ☒ 4. 750 M

Question ID : 8401605802
 Status : Answered
 Chosen Option : 2

Q.25 Statements: $P = Q \geq R = S$, $T > U \geq V = S$

Conclusions: I. $U \geq P$ II. $P \geq V$ III. $T \geq Q$ IV. $T > R$

Which one of the above conclusions is correct?

- Ans
- ☒ 1. Only I and III are true
 - ☒ 2. Only II and IV are true
 - ☒ 3. All I, II, III and IV are true
 - ☒ 4. Only II is true

Question ID : 8401605792
 Status : Answered
 Chosen Option : 1

Q.26 Indicating towards a photograph of a male child, Rohit told to Amit that the child in the photograph is the son of the only son of my mother. How is Rohit related to the male child in the photograph?

- Ans
- ☒ 1. Father
 - ☒ 2. Uncle
 - ☒ 3. Son
 - ☒ 4. Grand Father

Question ID : 8401605803
 Status : Answered
 Chosen Option : 2

Q.27 If RAID is coded as 6821, THAT is coded as 7387 and PUT is coded as 457, what will be the code for RAT?

- Ans
- ☒ 1. 786
 - ☒ 2. 887
 - ☒ 3. 687
 - ☒ 4. 867

Question ID : 8401605798
Status : Answered
Chosen Option : 3

Q.28 Rohit is in the East of Mohan which is in the North of Sushil. If Punit is in the South of Sushil, then in which direction of Rohit, is Punit?

- Ans
- ☒ 1. South-West
 - ☒ 2. North-West
 - ☒ 3. North-East
 - ☒ 4. South-East

Question ID : 8401605800
Status : Answered
Chosen Option : 3

Q.29 If the day before yesterday was Friday, when will Tuesday be?

- Ans
- ☒ 1. Two days after tomorrow
 - ☒ 2. Day after tomorrow
 - ☒ 3. Today
 - ☒ 4. Tomorrow

Question ID : 8401605786
Status : Answered
Chosen Option : 4

Q.30 There are seven boxes named Mega, Meta, Rita, Docu, Gimu, Tina, Pina and Lopu of different colours, like; Black, Violet, Red, Blue, Yellow, White and Green but not necessarily in the same order. Only two boxes are kept between yellow colour box and box Docu, which is kept above yellow colour box. Box Docu is not of green colour. Black colour box is kept above violet colour box. Neither Box Rita nor Box Gimu is of Black colour. Box Gimu is kept above box Tina. There are two boxes between box Mega and the box which is Green in colour. Box Rita is kept either immediately above or immediately below green colour box. More than two boxes are kept between Box Rita and the box which is of red colour, which is neither kept at the top nor at the bottom. Box Mega is not of red colour. Only one box is kept between red colour box and the white colour box. Box Mega and box Gimu is not of white colour. Box Meta is of blue colour and is not kept at the top and not at the bottom.

Which box is kept immediately on the top of Box Docu?

- Ans
- ☒ 1. Tina
 - ☒ 2. Rita
 - ☒ 3. Mega
 - ☒ 4. Meta

Question ID : 8401605787
Status : Answered
Chosen Option : 3

Q.31 In the given question sets of alphabets are given as options and these sets shares a common similarity, whereas one is different. Choose the odd one out.

- Ans ☒ 1. QIB
☒ 2. XUH
☒ 3. ACV
☒ 4. SMO

Question ID : 8401605808
 Status : Answered
 Chosen Option : 3

Q.32 Following table gives the details of sales of different items sold by three different stores run by three women. Please read the table carefully and answer the question given at the end of table.

Type / Sale in Thousand	Ambika Stores	Mahima Stores	Rohini Stores
Watches	312	234	453
Calculators	231	211	342
Mob. Phones	456	765	889
Coffee Mugs	2109	2306	3212

Find the difference of total number of all the items sold by the Rohini and Ambika Stores. Mugs.

- Ans ☒ 1. 1303
☒ 2. 1723
☒ 3. 1803
☒ 4. 1788

Question ID : 8401605814
 Status : Answered
 Chosen Option : 1

Q.33 Amar, Beerbhan, Vaishali, Mohit, Nitin, Pallavi, Jyoti, Preeti and Sonu are nine friends and they are sitting around a circular table by facing the centre of the table. Mohit is sitting second to the right of Pallavi. Only two people are sitting between Mohit and Vaishali. Two people are sitting between Nitin and Amar. Neither Preeti nor Nitin are an immediate neighbor of Vaishali or Mohit. Only one person is sitting between Vaishali and Nitin. Preeti is an immediate neighbor of Nitin. Amar sits third to the left of Jyoti. Vaishali sits second to the right of Beerbhan.

Who is sitting on the immediate left of Vaishali?

- Ans ☒ 1. Mohit
☒ 2. Jyoti
☒ 3. Amar
☒ 4. Nitin

Question ID : 8401605783
 Status : Answered
 Chosen Option : 2

Q.34 Choose the correct alternative from the given options which will continue the same pattern and replace the question mark in the given number series.

1, 2, 5, 26, 677,?

- Ans
- ☒ 1. 330458
 - ☒ 2. 485033
 - ☒ 3. 458330
 - ☒ 4. 330584

Question ID : 8401605794
Status : Answered
Chosen Option : 2

Section : Domain Knowledge

Q.1 If the controlling force equation of a spring-controlled governor is given by $F = p \cdot r + q$ (where r is the radius of rotation of governor balls), then the governor is:

- Ans
- ☒ 1. insensitive
 - ☒ 2. stable
 - ☒ 3. unstable
 - ☒ 4. isochronous

Question ID : 8401605833
Status : Answered
Chosen Option : 3

Q.2 Two springs, S1 and S2, with mean diameters of 90 mm and 75 mm, respectively, are made up of two equal lengths of hardened steel wires of the same diameter. The ratio of the stiffness of S1 to that of S2 is:

- Ans
- ☒ 1. 125/216
 - ☒ 2. 25/36
 - ☒ 3. 216/125
 - ☒ 4. 36/25

Question ID : 8401605849
Status : Answered
Chosen Option : 3

Q.3 The absolute maximum value of swaying couple is achieved when θ is:

- Ans
- ☒ 1. 45° and 225°
 - ☒ 2. 90° and 270°
 - ☒ 3. 0° and 180°
 - ☒ 4. 45° and 135°

Question ID : 8401605835
Status : Answered
Chosen Option : 4

Q.4 A thermal energy reservoir at 1200 K supplies 500 kJ of heat to a reversible heat engine E1. Engine E1 rejects heat Q_2 to a reversible heat engine E2 at temperature T_2 . Engine E2 rejects heat to a thermal energy reservoir at temperature 300 K. If the efficiency of both the engines is the same, then $T_1 : T_2 : T_3$ is:

- Ans
- ☒ 1. 2 : 1 : 1.414
 - ☒ 2. 2 : 1.5 : 1
 - ☒ 3. 2 : 1 : 0.5
 - ☒ 4. 2 : 1.414 : 1

Question ID : 8401605875
Status : Answered
Chosen Option : 3

Q.5 In soldering, the metal parts are joined by means of a fusible alloy which is composed of:

- Ans
- ☒ 1. Lead and Tin
 - ☒ 2. Lead and Copper
 - ☒ 3. Tin and Aluminium
 - ☒ 4. Copper and Tin

Question ID : 8401605892
Status : Answered
Chosen Option : 1

Q.6 A car of 1200 kg is moving on a straight level road with the speed of 10 m/s when the traffic lights turn from green to amber. The driver applies the breaks 20 m before the lights and just manages to stop on the line. The force applied to stop the car is:

- Ans
- ☒ 1. 2250 N
 - ☒ 2. 2750 N
 - ☒ 3. 3000 N
 - ☒ 4. 1500 N

Question ID : 8401605820
Status : Answered
Chosen Option : 3

Q.7 If the average arrival rate in a queue is 13/hr and the average service rate is 20/hr, then the average number of customers in the line (including the customer being served) will be:

- Ans
- ☒ 1. 1.857
 - ☒ 2. 1.908
 - ☒ 3. 1.946
 - ☒ 4. 1.880

Question ID : 8401605898
Status : Answered
Chosen Option : 3

Q.8 A square beam is placed in two arrangements:

- (i) with its side horizontal, and
- (ii) with a diagonal horizontal.

For a given stress, the beam in arrangement _____.

- Ans
- ☒ 1. (ii) is 41.4% more stronger than in (i)
 - ☒ 2. (ii) is 67.3% more stronger than in (i)
 - ☒ 3. Both arrangements have equal strength
 - ☒ 4. (i) is 41.4% more stronger than in (ii)

Question ID : 8401605821
Status : Answered
Chosen Option : 1

Q.9 Experimental values of critical compressibility factor (Z_c) for most gases fall in a narrow range of:

- Ans
- ☒ 1. 0.1 to 0.2
 - ☒ 2. 0.01 to 0.1
 - ☒ 3. 0.2 to 0.3
 - ☒ 4. 0.3 to 1.0

Question ID : 8401605874
Status : Answered
Chosen Option : 2

Q.10 Which of the following mechanisms converts rotary motion to sliding motion?

- Ans
- ☒ 1. Elliptical trammel
 - ☒ 2. Whitworth mechanism
 - ☒ 3. Scotch yoke mechanism
 - ☒ 4. Geneva mechanism

Question ID : 8401605828
Status : Answered
Chosen Option : 2

Q.11 Consider the following statements regarding heat treatment processes:

- 1. Tempering is a process of annealing bainite at low temperatures.
- 2. Machine tool guideways are hardened by flame hardenings
- 3. The aim of full annealing is to reduce ductility and resilience.
- 4. Guideways of lathe beds are hardened by carburising.

Which of the given statements are INCORRECT?

- Ans
- ☒ 1. 1, 2 and 4
 - ☒ 2. 1, 3 and 4
 - ☒ 3. 1, 2 and 3
 - ☒ 4. 2, 3 and 4

Question ID : 8401605880
Status : Answered
Chosen Option : 3

Q.12 A metallic body floats at the interface of mercury of specific gravity of 13.6 and water such that 40% of its volume is submerged in mercury and 60% in water. The density (kg/m^3) of the metallic body is approximately _____.

- Ans ☒ 1. 6040
☒ 2. 5580
☒ 3. 5720
☒ 4. 6230

Question ID : 8401605860
 Status : Answered
 Chosen Option : 3

Q.13 "If a body is in equilibrium under the action of three forces, then each force is proportional to the sine of the angle between the other two forces." This statement is known as:

- Ans ☒ 1. Coriolis Law
☒ 2. Pascal's Law
☒ 3. Lami's Theorem
☒ 4. D'Alembert's Principle

Question ID : 8401605819
 Status : Answered
 Chosen Option : 3

Q.14 Consider the following statements about Wein's displacement law:

1. Maximum spectral emissive power is displaced to longer wavelengths with increase in temperature.
2. Maximum spectral emissive power increases with decrease in temperature.
3. Maximum spectral emissive power is displaced to shorter wavelengths with increase in temperature.
4. Maximum spectral emissive power decreases with decrease in temperature.

Which of the given statements are correct?

- Ans ☒ 1. 3 and 4
☒ 2. 1 and 2
☒ 3. 1 and 3
☒ 4. 2 and 3

Question ID : 8401605869
 Status : Answered
 Chosen Option : 1

Q.15 An axial force that ranges from 50 kN in compression to 150 kN in tension is applied at the centre of a forged steel rod with a uniform diameter of 30 mm. Assume that the material has a tensile, yield and endurance strength of 600, 420 and 240 MPa, respectively. In accordance with Soderberg's criterion, the factor of safety is:

- Ans ☒ 1. 2.80
☒ 2. 1.36
☒ 3. 1.53
☒ 4. 2.25

Question ID : 8401605850
 Status : Answered
 Chosen Option : 3

Q.16 Lumped heat capacity analysis is valid for bodies for which the value of _____.

- Ans
- ☒ 1. $Bi > 0.1$
 - ☒ 2. $Bi < 0.1$
 - ☒ 3. $Fo < 0.1$
 - ☒ 4. $Bi < 1$

Question ID : 8401605864
Status : Answered
Chosen Option : 2

Q.17 Which of the following statements is correct regarding the term 'Reduced Pressure'?

- Ans
- ☒ 1. It is represented by the ordinate (Y axis) of generalised compressibility chart.
 - ☒ 2. It is the difference between the critical pressure and the existing pressure of a substance.
 - ☒ 3. It is the ratio of the existing pressure to the critical pressure of a substance.
 - ☒ 4. Its unit is MPa.

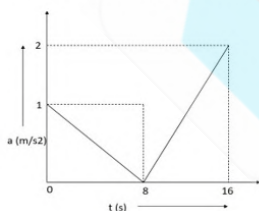
Question ID : 8401605872
Status : Answered
Chosen Option : 4

Q.18 The 'longitudinal bar stays' of a boiler are pitched at 0.35 m horizontally and vertically. The steam pressure is 10 bar. If the tensile stress of the material is 60 N/mm^2 , then the core diameter will be close to:

- Ans
- ☒ 1. 45 mm
 - ☒ 2. 60 mm
 - ☒ 3. 51 mm
 - ☒ 4. 56 mm

Question ID : 8401605843
Status : Answered
Chosen Option : 1

Q.19 An acceleration-time diagram of motion of a particle is shown in the given figure. If the initial velocity of the particle is 8 m/s, then the displacement of the particle till the end of 16th second is:



- Ans
- ☒ 1. 117.33 m
 - ☒ 2. 202.63 m
 - ☒ 3. 227.80 m
 - ☒ 4. 192.15 m

Question ID : 8401605817
Status : Answered
Chosen Option : 3

Q.20 If Moment of inertia, rotational kinetic energy, and angular momentum is given by M, K, and L, then:

Ans ☒ 1. $K = 2LM$

☒ 2. $L = \sqrt{(2MK)}$

☒ 3. $M = K/L$

☒ 4. $L = MK$

Question ID : 8401605901
Status : Answered
Chosen Option : 1

Q.21 The water velocity in a horizontal pipe with a 30 mm diameter is measured using a venturi-meter with a 15 mm throat diameter. When frictional losses are neglected and the pressure difference between the pipe and the throat sections is 20 kPa, the flow velocity is:

Ans ☒ 1. 1.93 m/s

☒ 2. 1.63 m/s

☒ 3. 2.0 m/s

☒ 4. 1.45 m/s

Question ID : 8401605858
Status : Answered
Chosen Option : 2

Q.22 Column A has both its ends fixed, and column B has one end fixed and the other end free. The ratio of the buckling load of column A to that of column B is:

Ans ☒ 1. 1

☒ 2. 16

☒ 3. 0.5

☒ 4. 2

Question ID : 8401605822
Status : Answered
Chosen Option : 2

Q.23 A Hartnell governor is a/an _____ governor.

Ans ☒ 1. pendulum type

☒ 2. inertia type

☒ 3. spring-loaded type

☒ 4. dead weight type

Question ID : 8401605832
Status : Answered
Chosen Option : 3



Q.24 Consider the following statements regarding limit gauges:

1. Plug gauge is used to check hole size.
2. Ring gauge is used to measure the external diameter of a shaft.
3. Both ring and plug gauges are available in two designs, 'GO' and 'NO-GO'.

Which of the given statements are correct?

- Ans
- ☒ 1. 1 and 3
 - ☒ 2. 1, 2 and 3
 - ☐ 3. 2 and 3
 - ☐ 4. 1 and 2

Question ID : 8401605894
Status : Answered
Chosen Option : 4

Q.25 The maximum permissible twisting moment in a circular shaft, according to the maximum shear stress theory of failure, is 'T'. According to the maximum principal stress theory of failure, the permissible twisting moment for the same shaft is:

- Ans
- ☒ 1. $2T$
 - ☐ 2. $\sqrt{2} T$
 - ☐ 3. T
 - ☐ 4. $\frac{T}{2}$

Question ID : 8401605848
Status : Answered
Chosen Option : 2

Q.26 Consider the following statements regarding Heisler Chart:

1. Heisler Chart is used to determine temperature distribution and transient heat flow rate.
2. Heisler chart is used to determine temperature distribution and steady heat flow rate.
3. Heisler Chart is used when conduction and convection resistances are almost of equal importance.
4. Heisler Chart is used when conduction resistance is higher than convection resistance.

Which of the given statements are correct?

- Ans
- ☒ 1. 1 and 3
 - ☐ 2. 2 and 3
 - ☐ 3. 1 and 4
 - ☐ 4. 1, 3 and 4

Question ID : 8401605863
Status : Answered
Chosen Option : 3



Q.27 A rectangular sluice gate is installed on the vertical wall of a lock. If the vertical side of gate is 'd' metres and its centroid is 'p' metres below the water surface, then the depth of pressure is:

- Ans
- ✓ 1. $p + (d^2/12p)$
 - ✗ 2. $p + (d^2/6p)$
 - ✗ 3. $p + (d^2/p)$
 - ✗ 4. $p + (3d^2/4p)$

Question ID : 8401605852
Status : Answered
Chosen Option : 1

Q.28 Consider the following statements regarding project management:

1. PERT is probabilistic and CPM is deterministic.
2. In PERT, network slack on various events is calculated, whereas in CPM, floats are calculated.
3. In PERT, activity duration is normally distributed and project duration is beta distributed.
4. The critical path of a network is the path that takes the shortest time.

Which of the given statements are INCORRECT?

- Ans
- ✗ 1. 1 and 2
 - ✗ 2. 1, 3 and 4
 - ✓ 3. 3 and 4
 - ✗ 4. 2 and 3

Question ID : 8401605899
Status : Answered
Chosen Option : 3

Q.29 Entropy generation number is mathematically expressed as:

- Ans
- ✗ 1. $\frac{\dot{S}_{gen}}{\dot{m} R}$
 - ✓ 2. $\frac{\dot{S}_{gen}}{\dot{m} C_p}$
 - ✗ 3. $\frac{(C_p - C_v) \dot{m}}{\dot{S}_{gen}}$
 - ✗ 4. $\frac{C_p \dot{m}}{\dot{S}_{gen}}$

Question ID : 8401605876
Status : Answered
Chosen Option : 2



Q.30 Consider the following statements regarding the effect of alloying elements in steel:

1. Sulphur - Improves machinability in free cutting steel
2. Molybdenum - Forms abrasion resisting particles
3. Phosphorous - Improves Corrosion resistance
4. Vanadium - Increases fatigue strength

Which of the given statements are INCORRECT?

- Ans ☒ 1. 3 and 4
- ☒ 2. 2 and 3
- ☒ 3. 1 and 4
- ☒ 4. 1 and 2

Question ID : 8401605886
Status : Answered
Chosen Option : 1

Q.31 A spring mass system consists of a mass of 5 kg and two springs of stiffness 8 N/mm and 12 N/mm. The system is arranged in different manners, that is:

- (i) the mass is suspended at the bottom of two springs in series, and
- (ii) the mass is fixed between two springs.

The ratio of natural frequencies of case (ii) to those of case (i) is approximately _____.

- Ans ☒ 1. 0.5
- ☒ 2. 1.0
- ☒ 3. 2.0
- ☒ 4. 1.5

Question ID : 8401605840
Status : Answered
Chosen Option : 3

Q.32 Which of the following is the correct order of cooling media for increasing cooling rate?

- Ans ☒ 1. Water, Air, Fused salt and Oil
- ☒ 2. Air, Fused salt, Oil and Water
- ☒ 3. Oil, Fused salt, Air and Water
- ☒ 4. Fused salt, Oil, Water and Air

Question ID : 8401605879
Status : Answered
Chosen Option : 2

Q.33 The three-time estimates of a PERT activity, i.e. optimistic time, pessimistic time and most likely time, are 7 min, 13 min and 9 min, respectively. The expected time (in minutes) and the variance of the activity will be:

- Ans ☒ 1. 15.67 and 1.5
- ☒ 2. 11.33 and 1.5
- ☒ 3. 15.67 and 1
- ☒ 4. 11.33 and 1

Question ID : 8401605900
Status : Answered
Chosen Option : 2

Q.34 Consider the following statements regarding the stress strain diagram:

1. Hooke's law is followed up to the elastic limit.
2. The stress at which a material elongates without an increase in load is called as yield stress.
3. Toughness is usually represented by the area under a stress strain curve.

Which of the given statements are correct?

- Ans
- ☒ 1. 1 and 3
 - ☒ 2. Only 1
 - ☒ 3. 1 and 2
 - ☒ 4. 2 and 3

Question ID : 8401605884
Status : Answered
Chosen Option : 1

Q.35 A pipe of diameter 100 mm is suddenly enlarged to a diameter of 200 mm. If the flow rate of water is 200 litres/s, then the loss of head is:

- Ans
- ☒ 1. 3.25 m of water
 - ☒ 2. 5.21 m of water
 - ☒ 3. 4.64 m of water
 - ☒ 4. 4.15 m of water

Question ID : 8401605856
Status : Answered
Chosen Option : 3

Q.36 Which of the following statements is correct?

- Ans
- ☒ 1. TTT and Fe-Fe₃C diagrams are both non-equilibrium diagrams.
 - ☒ 2. TTT diagram is an equilibrium diagram, but Fe-Fe₃C diagram is a non-equilibrium diagram.
 - ☒ 3. TTT and Fe-Fe₃C diagrams are both equilibrium diagrams.
 - ☒ 4. TTT diagram is a non-equilibrium diagram, but Fe-Fe₃C diagram is an equilibrium diagram.

Question ID : 8401605883
Status : Answered
Chosen Option : 2

Q.37 200 kJ of energy is transferred from a heat reservoir at 1050 K to a heat reservoir at 550 K. The ambient temperature is 310 K. The loss of available energy due to heat transfer process is:

- Ans
- ☒ 1. 59.28 kJ
 - ☒ 2. 50.67 kJ
 - ☒ 3. 53.68 kJ
 - ☒ 4. 55.33 kJ

Question ID : 8401605878
Status : Answered
Chosen Option : 2



Q.38 A vibrating system having a mass of 30 kg is supported by 4 springs, each having a stiffness of k. The system runs at 420 rpm. If only 10% of the shaking force is allowed to be transmitted to the supporting structure, then the value of stiffness k will be close to:

- Ans
- ☒ 1. 1.09 N/mm
 - ☒ 2. 3.96 N/mm
 - ☒ 3. 2.18 N/mm
 - ☒ 4. 1.32 N/mm

Question ID : 8401605837
Status : Answered
Chosen Option : 3

Q.39 Which of the following heat exchangers gives parallel and linear temperature profile for both cold and hot fluid?

- Ans
- ☒ 1. Parallel flow with unequal heat capacities
 - ☒ 2. Counter flow with equal heat capacities
 - ☒ 3. Parallel flow with equal heat capacities
 - ☒ 4. Crossed flow with unequal heat capacities

Question ID : 8401605865
Status : Answered
Chosen Option : 4

Q.40 In a counter flow heat exchanger, if the hot and cold fluids enter at T₁ and T₂, respectively, whereas cold fluid leaves at T₃ and hot fluid leaves at T₄, then LMTD is given by:

- Ans
- ☒ 1. $\frac{(T_1 - T_3) - (T_2 - T_4)}{\ln \frac{(T_1 - T_3)}{(T_2 - T_4)}}$
 - ☒ 2. $\frac{(T_1 - T_2) + (T_3 - T_4)}{\ln \frac{(T_1 - T_2)}{(T_3 - T_4)}}$
 - ☒ 3. $\frac{(T_1 - T_2) - (T_3 - T_4)}{\ln \frac{(T_1 - T_2)}{(T_3 - T_4)}}$
 - ☒ 4. $\frac{(T_1 - T_3) - (T_4 - T_2)}{\ln \frac{(T_1 - T_3)}{(T_4 - T_2)}}$

Question ID : 8401605866
Status : Answered
Chosen Option : 3



Q.41 Consider the following statements about the thermal diffusivity of a substance:

1. It is directly proportional to density.
 2. It is inversely proportional to specific heat.
 3. It is inversely proportional to thermal conductivity.
 4. It is directly proportional to thermal conductivity.
- Which of the given statements are INCORRECT?

Ans ☒ 1. 1, 2 and 3
☒ 2. 2 and 3
☒ 3. 1 and 2
☒ 4. 1 and 3

Question ID : 8401605861
 Status : Answered
 Chosen Option : 3

Q.42 In a slider-crank mechanism, the velocity of piston becomes maximum when:

Ans ☒ 1. the crank is at an angle of 120° with the line of stroke
☒ 2. the crank is perpendicular to the line of stroke of the piston
☒ 3. the crank and the connecting rod are mutually perpendicular
☒ 4. the crank and the connecting rod are in line with each other

Question ID : 8401605829
 Status : Answered
 Chosen Option : 2

Q.43 A two-dimensional fluid element rotates like a rigid body, and at a point within the element, the pressure is 1 unit. What is the radius of Mohr's circle, characterising the state of stress at that point?

Ans ☒ 1. 0.5 unit
☒ 2. 0 unit
☒ 3. 1 unit
☒ 4. 2 units

Question ID : 8401605823
 Status : Answered
 Chosen Option : 2



Q.44 Consider the following statements regarding linear programming:

1. A linear programming problem with three variables and two constraints cannot be solved by the graphical method.
2. In the Big-M method, when the artificial variable leaves the basis, its column can be deleted from the subsequent tables.
3. When one constraint line comes parallel to the objective function line, LPP will have a degenerate solution.

Which of the given statements are correct?

- Ans
- ☒ 1. 2 and 3
 - ☒ 2. 1 and 2
 - ☒ 3. 1 and 3
 - ☒ 4. 1, 2 and 3

Question ID : 8401605896
Status : Answered
Chosen Option : 2

Q.45 The velocity components in the x and y directions are given by:

$$u = \lambda \times x \times (y^2) - (x^2) \times y \text{ and } v = x \times (y^2) - \frac{2}{3} \times (y^3)$$

For a continuous flow, the value of λ is:

- Ans
- ☒ 1. $\frac{2}{3}$
 - ☒ 2. 2
 - ☒ 3. $\frac{3}{2}$
 - ☒ 4. $-\frac{3}{2}$

Question ID : 8401605853
Status : Answered
Chosen Option : 1

Q.46 In the boundary layer, the flow is:

- Ans
- ☒ 1. inviscid and rotational
 - ☒ 2. viscous and rotational
 - ☒ 3. viscous and irrotational
 - ☒ 4. inviscid and irrotational

Question ID : 8401605855
Status : Answered
Chosen Option : 2



Q.47 A machine element is subjected to the following bi-axial state of stress:

$$\sigma_x = 70 \text{ MPa}; \sigma_y = 30 \text{ MPa}; \tau_{xy} = 40 \text{ MPa}.$$

If the shear strength of the material is 100 MPa, then the factor of safety as per Tresca's theory is:

- Ans
- ☒ 1. 3.83
 - ☒ 2. 2.23
 - ☒ 3. 3.15
 - ☒ 4. 4.07

Question ID : 8401605847
Status : Answered
Chosen Option : 2

Q.48 The 'momentum correction factor' for a laminar flow through a circular pipe is:

- Ans
- ☒ 1. $\frac{1}{3}$
 - ☒ 2. $\frac{2}{3}$
 - ☒ 3. $\frac{3}{4}$
 - ☒ 4. $\frac{4}{3}$

Question ID : 8401605854
Status : Answered
Chosen Option : 2

Q.49 Which of the following components can be made by powder metallurgy technique?

1. Porous bearing
2. Filters
3. Gears
4. Turbine blades

- Ans
- ☒ 1. Only 1 and 2
 - ☒ 2. 1, 2 and 4
 - ☒ 3. 1, 2 and 3
 - ☒ 4. 2, 3 and 4



Question ID : 8401605891
Status : Answered
Chosen Option : 3



Q.50 The actual demand of a product in a company is 79 units. Previous forecast and exponential smoothening factor are 84 units and 0.25, respectively. What will be the forecast (of the product units) for the next period?

- Ans
- ☒ 1. 81.75
 - ☒ 2. 80.25
 - ☒ 3. 82.75
 - ☒ 4. 81.25

Question ID : 8401605895
Status : Answered
Chosen Option : 3

Q.51 How many occupied cells must a transportation matrix with 8 rows and 7 columns have so that it DOES NOT degenerate?

- Ans
- ☒ 1. 15
 - ☒ 2. 55
 - ☒ 3. 56
 - ☒ 4. 14

Question ID : 8401605897
Status : Answered
Chosen Option : 2

Q.52 Which of the following are the features of submerged arc welding?

1. High welding speeds
2. Less radiation hazard
3. Low deposition rate
4. High quality of Weld

- Ans
- ☒ 1. 2, 3 and 4
 - ☒ 2. 1, 2 and 4
 - ☒ 3. 1, 2 and 3
 - ☒ 4. Only 1 and 3

Question ID : 8401605893
Status : Answered
Chosen Option : 1

Q.53 Which of the following processes are irreversible?

1. Free expansion
2. Transfer of electricity through a resistor
3. Heat transfer through a finite temperature difference
4. Isothermal expansion in a Stirling cycle
5. Mixing of two ideal gases at constant pressure

- Ans
- ☒ 1. 1, 2, 3 and 5
 - ☒ 2. 1, 2, 3 and 4
 - ☒ 3. 1, 3, 4 and 5
 - ☒ 4. 2, 3, 4 and 5

Question ID : 8401605877
Status : Answered
Chosen Option : 3

Q.54 A worm wheel has 40 teeth and a pitch diameter of 240 mm. If the pressure angle of the worm is 20° , then the axial pitch of the worm will be close to:

- Ans ☒ 1. 18.85 mm
☒ 2. 22 mm
☒ 3. 31 mm
☒ 4. 25.14 mm

Question ID : 8401605831
 Status : Answered
 Chosen Option : 3

Q.55 For real fluids, the 'loss of head' term is incorporated in Bernoulli's equation. This term is associated with fluid's _____.

- Ans ☒ 1. compressibility
☒ 2. temperature
☒ 3. viscosity
☒ 4. surface tension

Question ID : 8401605857
 Status : Answered
 Chosen Option : 3

Q.56 A strip is to be rolled from a thickness of 36 mm to 20 mm using a two-high mill having rolls of diameter 400 mm. The approximate value of the coefficient of friction for unaided bite will be:

- Ans ☒ 1. 0.22
☒ 2. 0.25
☒ 3. 0.28
☒ 4. 0.20

Question ID : 8401605890
 Status : Answered
 Chosen Option : 1

Q.57 In a real gas equation $pV = ZRT$, depending on the values of temperature and pressure of the real gas, the value of Z _____.

- Ans ☒ 1. should always be greater than unity
☒ 2. may be less than, greater than or equal to unity
☒ 3. should always be equal to unity
☒ 4. should always be less than unity

Question ID : 8401605871
 Status : Answered
 Chosen Option : 2

Q.58 Which of the following is NOT a pure substance?

- Ans ☒ 1. Mixture of air (gas phase) and liquid air
☒ 2. Combustion product of a fuel
☒ 3. Atmospheric air
☒ 4. Vapours of ammonia

Question ID : 8401605870
 Status : Answered
 Chosen Option : 2

Q.59 Charpy test is used to measure:

- Ans ☒ 1. tensile strength
☒ 2. impact strength
☒ 3. bending strength
☒ 4. fatigue strength

Question ID : 8401605825
 Status : Answered
 Chosen Option : 2

Q.60 Which of the following statements about contact ratio is correct?

- Ans ☒ 1. Both addendum and base circle diameter have influence on contact ratio.
☒ 2. Addendum has no effect on contact ratio.
☒ 3. Increasing the addendum results in a higher value of contact ratio.
☒ 4. Decreasing the addendum results in a higher value of contact ratio.

Question ID : 8401605830
 Status : Answered
 Chosen Option : 3

Q.61 In an infinitely long fin, if the parameter $m = \sqrt{\frac{hP}{kA}}$ increases, the other parameters being maintained constant, then _____.

- Ans ☒ 1. the temperature drop along the length will be steeper
☒ 2. the temperature profile will remain the same
☒ 3. the temperature drop along the length will be at a low rate
☒ 4. only the heat flow rate will be increased without any effect on the temperature drop

Question ID : 8401605862
 Status : Answered
 Chosen Option : 4

Q.62 A spherical drop of molten metal of radius 3 mm was found to solidify in 12 seconds. A similar drop of radius 7 mm would solidify in:

- Ans
- ☒ 1. 61.74 seconds
 - ☒ 2. 28 seconds
 - ☒ 3. 65.33 seconds
 - ☒ 4. 63.22 seconds

Question ID : 8401605889
Status : Answered
Chosen Option : 2

Q.63 NTU effectiveness method for the analysis of heat exchanger is used when:

- Ans
- ☒ 1. outlet temperatures of both fluids are known but inlet temperatures are unknown
 - ☒ 2. inlet temperatures of both fluids are known but outlet temperatures are unknown
 - ☒ 3. outlet temperatures of any one fluid is known
 - ☒ 4. inlet temperatures of any one fluid is known

Question ID : 8401605867
Status : Answered
Chosen Option : 1

Q.64 Consider a uniform shaft of length L fixed at its upper end and carrying a disc of moment of inertia I at its lower end. The disc is twisted about the vertical axis and released. ' f_a ' is the natural frequency of the system when the shaft is assumed as massless, and ' f_b ' is the natural frequency of the system when the shaft is considered of the same moment of inertia as that of the disc. Find the ratio f_a/f_b .

- Ans
- ☒ 1. $\sqrt{4/3}$
 - ☒ 2. $\sqrt{3/4}$
 - ☒ 3. $3/4$
 - ☒ 4. $4/3$

Question ID : 8401605841
Status : Answered
Chosen Option : 4

Q.65 In which of the following processes is the Widmanstätten structure encountered?

- Ans
- ☒ 1. Tempering
 - ☒ 2. Nitriding
 - ☒ 3. Cyaniding
 - ☒ 4. Annealing

Question ID : 8401605881
Status : Answered
Chosen Option : 2



Q.66 Consider the following statements about determining the natural frequency of a shaft carrying several loads. Identify the correct option.

- Ans ☒ 1. Dunkerley's method is semi-empirical and simple.
- ☒ 2. Dunkerley's method provides accurate results.
- ☒ 3. The energy method involves fewer calculations in comparison to Dunkerley's method.
- ☒ 4. The energy method provides approximate results.

Question ID : 8401605839
Status : Answered
Chosen Option : 2

Q.67 Consider the following regarding crystal structure:

1. Simple Cubic - Manganese
2. Body-Centred Cubic - Copper
3. Face-Centred Cubic - Alpha Iron at room temperature
4. Hexagonal Close Packed - Zinc

Which of the above matches are correct?

- Ans ☒ 1. 2 and 3
- ☒ 2. 1 and 4
- ☒ 3. 3 and 4
- ☒ 4. 1 and 2

Question ID : 8401605887
Status : Answered
Chosen Option : 2

Q.68 Real gas behaves like an ideal gas when:

- Ans ☒ 1. pressure approaches 100 kPa and temperature approaches infinity
- ☒ 2. pressure approaches zero and temperature approaches infinite
- ☒ 3. both pressure and temperature approach infinite
- ☒ 4. pressure approaches infinite and temperature approaches 300 K

Question ID : 8401605873
Status : Answered
Chosen Option : 2

Q.69 In order to avoid seizure in hydrodynamic bearings, the operating value of the bearing characteristic number should be at least ____ times the bearing modulus.

- Ans ☒ 1. 10
- ☒ 2. 5
- ☒ 3. 15
- ☒ 4. 20

Question ID : 8401605846
Status : Answered
Chosen Option : 3



Q.70 A ball-bearing is characterised by a basic static capacity of 12000 N and a dynamic capacity of 17000 N. This bearing is subjected to an equivalent static load of 5000 N. The bearing loading ratio and life in million revolutions, respectively, are:

- Ans
- ☒ 1. 3.81 and 54.87
 - ☒ 2. 2.4 and 13.82
 - ☒ 3. 3.4 and 39.30
 - ☒ 4. 3.27 and 34.96

Question ID : 8401605844
Status : Answered
Chosen Option : 2

Q.71 A suspended mass of 10 kg completes 40 oscillations in 20 seconds in a single-degree damped vibrating system. The stiffness of the spring is approximately _____.

- Ans
- ☒ 1. 2.24 N/mm
 - ☒ 2. 1.12 N/mm
 - ☒ 3. 0.79 N/mm
 - ☒ 4. 1.58 N/mm

Question ID : 8401605838
Status : Answered
Chosen Option : 4

Q.72 Which of the following phases of steel is NOT present in the Fe-Fe₃C phase diagram?

- Ans
- ☒ 1. Ferrite
 - ☒ 2. Cementite
 - ☒ 3. Austenite
 - ☒ 4. Martensite

Question ID : 8401605882
Status : Answered
Chosen Option : 1

Q.73 Two pipes, each of diameter 5 cm and 15 cm, are used to drain water from a reservoir. If the frictional head losses in both pipes are equal, then the ratio of discharge through the larger pipe to that through the smaller pipe will be:

- Ans
- ☒ 1. 9
 - ☒ 2. $\sqrt{3}$
 - ☒ 3. $6\sqrt{3}$
 - ☒ 4. $9\sqrt{3}$

Question ID : 8401605859
Status : Answered
Chosen Option : 4



Q.74 For a single degree of freedom viscous damped system, if the frequency ratio is greater than $\sqrt{2}$, it implies that:

- Ans
- ☒ 1. the transmitted force is more than the exciting force
 - ☒ 2. the transmitted force is equal to the exciting force
 - ☒ 3. the transmitted force is less than the exciting force
 - ☒ 4. the transmitted force is infinite

Question ID : 8401605836
Status : Answered
Chosen Option : 1

Q.75 The whirling speed of a rotating shaft depends on its _____.

- Ans
- ☒ 1. mass and stiffness
 - ☒ 2. mass, stiffness and eccentricity
 - ☒ 3. mass and eccentricity
 - ☒ 4. Only stiffness

Question ID : 8401605842
Status : Answered
Chosen Option : 2

Q.76 Which of the following patterns is used for a casting where some portions of the casting are structurally weak and likely to break by the force of ramming?

- Ans
- ☒ 1. Follow board pattern
 - ☒ 2. Skeleton pattern
 - ☒ 3. Gated pattern
 - ☒ 4. Loose piece pattern

Question ID : 8401605888
Status : Answered
Chosen Option : 2

Q.77 A projectile is fired at an angle of 60° with the horizontal. If the horizontal range is 2 km, then the velocity of projection is approximately _____.

- Ans
- ☒ 1. 170 m/s
 - ☒ 2. 180 m/s
 - ☒ 3. 140 m/s
 - ☒ 4. 150 m/s

Question ID : 8401605818
Status : Answered
Chosen Option : 2



Q.78 A full journal bearing having clearance to radius ratio of $1/100$, using a lubricant with absolute viscosity $\mu = 28 \times 10^{-3}$ Pa.s, supports the shaft journal running at $N = 2400$ rpm. If the bearing pressure is 1.4 MPa, then the Sommerfeld number is:

- Ans
- ☒ 1. 0.48
 - ☒ 2. 8×10^{-3}
 - ☒ 3. 0.48×10^{-6}
 - ☒ 4. 8×10^{-5}

Question ID : 8401605845
Status : Answered
Chosen Option : 4

Q.79 The material medium between the heat source and the receiver is NOT affected during the process of heat transmission by _____.

- Ans
- ☒ 1. convection
 - ☒ 2. conduction as well as convection
 - ☒ 3. conduction
 - ☒ 4. radiation

Question ID : 8401605868
Status : Answered
Chosen Option : 1

Q.80 The most appropriate follower motion plan for high-speed engines is:

- Ans
- ☒ 1. cycloidal
 - ☒ 2. uniform velocity
 - ☒ 3. uniform acceleration and deceleration
 - ☒ 4. simple harmonic motion

Question ID : 8401605834
Status : Answered
Chosen Option : 1

Q.81 Which of the following hardness tests is best suitable for brittle materials such as ceramics?

- Ans
- ☒ 1. Brinell hardness test
 - ☒ 2. Knoop hardness test
 - ☒ 3. Vickers hardness test
 - ☒ 4. Rockwell hardness test

Question ID : 8401605826
Status : Answered
Chosen Option : 2

Q.82 If B is the centre of buoyancy, G is the centre of gravity and M is the metacentre of a floating body, then the body will be in stable equilibrium if _____.

- Ans
- ☐ 1. $BG = 0$
 - ☐ 2. $MG = 0$
 - ☐ 3. M is below G
 - ☒ 4. M is above G

Question ID : 8401605851
Status : Answered
Chosen Option : 4

Q.83 The number of instantaneous centres with 5 links is:

- Ans
- ☐ 1. 24
 - ☒ 2. 10
 - ☐ 3. 9
 - ☐ 4. 25

Question ID : 8401605827
Status : Answered
Chosen Option : 1

Q.84 Which of the following is the property of a material by virtue of which strain increases with time at constant load?

- Ans
- ☒ 1. Creeping
 - ☐ 2. Elasticity
 - ☐ 3. Malleability
 - ☐ 4. Yielding

Question ID : 8401605885
Status : Answered
Chosen Option : 1

Q.85 A metal bar of 50 mm × 50 mm section is subjected to a tensile load of 400 kN. The elongation of a 200 mm gauge length is found to be 0.2 mm, and the decrease in thickness is 0.012 mm. The value of Poisson's ratio is:

- Ans
- ☒ 1. 0.24
 - ☐ 2. 0.18
 - ☐ 3. 0.30
 - ☐ 4. 0.36

Question ID : 8401605824
Status : Answered
Chosen Option : 1



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